LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	27
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Deviation coefficients: - Coefficient A, Coefficient B, Coefficient C
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Deviation coefficients: - Coefficient A, Coefficient B, Coefficient C.
5. Specific Objectives	15.1.10.3 Trainee will be able to explain Deviation coefficients of earth magnetism.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of magnetic compass deviation cards; Instructor supervision required for safety regarding magnetic instruments.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary source of magnetic deviation in aircraft?
(b) How does the earth's magnetic field interact with aircraft metal?
(c) Why is it necessary to correct magnetic compass readings?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Deviation Coefficients • Role of Coefficient A • Role of Coefficient B • Role of Coefficient C • Mathematical representation of coefficients	L & Ds	PPT, IB	15 min
(b)	• Calculation of Coefficient A • Calculation of Coefficient B • Calculation of Coefficient C • Impact of aircraft heading on coefficients • Standardization of deviation values	L & Ds	PPT, PDF	20 min
(c)	• Application of coefficients in flight • Error analysis using coefficients • Correction procedures for pilots • Recording deviation data • Maintenance of deviation cards	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the significance of Coefficient A in magnetic deviation.
(b) How does Coefficient B affect compass readings at different headings?
(c) What steps are taken to correct for Coefficient C errors?

12. Home Assignment and Follow Up	Prepare a brief note on how deviation coefficients will be utilized in the next lesson regarding accelerometers and fatigue meters.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction – accelerometers and fatigue meters.	Refer to the master syllabus reference list	LP 28

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min